

Rudder Angle - Short Circuit (Short to Low)

Symptoms

- Alarm Code 7267 or 7664 displayed on ED-4 screen.
- Rudder angle is a constant value.

How To Use This Tree

This symptom tree can be used to troubleshoot rudder angle sensor alarm code. Start by performing Step 1 troubleshooting. Step 2 will ask a series of questions and will provide a list of troubleshooting steps to perform, depending on the symptom.

Shoptalk

Alarm code 7266 and 7265 are supported on ED-4 operating with any Software Version 5 or lower

Alarm code 7665 and 7664 are supported on ED-4 operating with any Software Version 6 or greater

Possible causes include:

- Rudder angle sensor malfunction
- Rudder angle signal shorted low.

Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Check the alarm codes.	
	STEP 1A. Check for sensor supply alarm codes.	Alarm Code 7267 or 7664 active?
STEP 2.	Check the ED-4 display.	
	STEP 2A. Verify sensor data in the ED-4 display.	Measured value matches the recorded value?
STEP 3.	Check the rudder angle level sensor and harness connector.	

STEPS		SPECIFICATIONS
	STEP 3A. Inspect the rudder angle level sensor and connector pins.	Dirty or damaged pins?
	STEP 3B. Check the circuit response.	Approximately 12 VDC?
	STEP 3C. Check the circuit response.	Alarm Code 7267 or 7664 active and Alarm Code 7267 or 7664 inactive?
	STEP 3D. Check the alarm codes and verify sensor condition.	Alarm Code 7267 or 7664 active?
STEP 4.	Check the original equipment manufacturer (OEM) sensor wiring harness.	
	STEP 4A. Inspect the OEM sensor wiring harness connector pins.	Dirty or damaged pins?
	STEP 4B. Check for a pin-to-pin short circuit in the OEM sensor wiring harness.	Greater than 100k ohms?
	STEP 4C. Check for a pin-to-ground short circuit.	Greater than 100k ohms?
	STEP 4D. Check for an inactive fault code.	Alarm Code 7267 or 7664 no longer active?

STEP 1. Check the alarm codes.

STEP 1A. Check for sensor supply alarm codes.

Conditions :

- Turn system enable switch ON.

Action	Specification/Repair	Next Step
• Wait 30 seconds. Check the ED-4 display to read the alarm codes.	Alarm Code 7267 or 7664 active?	2A
	YES	
	Alarm Code 7267 or 7664 active?	2A
	NO	

STEP 2. Check the ED-4 display.

STEP 2A. Verify sensor data in the ED-4 display.

Conditions :

- Turn system enable switch ON.

Action	Specification/Repair	Next Step
• Record the rudder angle sensor raw value in the I/O viewer. • Disconnect the ED-4 display from the Customer Interface Box (C.I.B.). Refer to Procedure 015-023 in section 15. (/qs3/pubsys2/xml/en/procedures/513/513-015-023.html) • Disconnect the primary and secondary connectors from the ED-4 display. • Place one lead on ED-4 display sensor SIGNAL pin 4 on the ED-4 secondary connector. • Place the other lead on the RETURN pin 1 on the ED-4 primary connector. Use the following procedure for general inspection techniques. Refer to Procedure 019-361 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-361.html)	Measured value matches the recorded value? YES	3A
	Measured value matches the recorded value? NO Repair : Check ED-4 display analog input channels in the data log and view - I/O viewer section. Refer to Procedure 015-035 in section 15. (/qs3/pubsys2/xml/en/procedures/513/513-015-035.html) Replace ED-4 if analog channel is not working properly. Refer to Procedure 015-023 in section 15. (/qs3/pubsys2/xml/en/procedures/513/513-015-023.html)	Repair complete.

STEP 3. Check the rudder angle level sensor and harness connector.

STEP 3A. Inspect the rudder angle level sensor and connector pins.

Conditions :

- Turn system enable switch OFF.
- Disconnect the rudder angle level sensor from the OEM sensor wiring harness.

Action	Specification/Repair	Next Step
• Loose connector • Corroded pins • Bent or broken pins • Pushed back or expanded pins • Moisture in or on the connector • Missing or damaged connector seals • Dirt or debris in or on the connector pins • Connector shell broken • Wire insulation damage • Damaged connector locking tab. Use the following procedure for general inspection techniques. Refer to Procedure 019-361 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-361.html)	Dirty or damaged pins? YES Repair : A damaged connection has been detected in the sensor or harness connector. Clean the connector and pins. Repair the damaged harness, connector, or pins. Replace the harness. Refer to Procedure 015-103 in Section 15. (/qs3/pubsys2/xml/en/procedures/513/513-015-103.html)	Repair complete.
	Dirty or damaged pins? NO	3B

STEP 3B. Check the circuit response.

Conditions :

- Turn system enable switch OFF.
- Disconnect the rudder angle sensor connector from the OEM sensor wiring harness.
- Turn system enable switch ON.

Action	Specification/Repair	Next Step
• Measure the voltage between the rudder angle level SUPPLY pin and the rudder angle level RETURN pin at the rudder angle level sensor connector of the OEM sensor wiring harness. Refer to the circuit diagram or wiring diagram for connector pin identification. Use the following procedure for general multimeter usage techniques. Refer to Procedure 019-359 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-359.html)	Approximately 12 VDC? YES	3C
	Approximately 12 VDC? NO	4A

STEP 3C. Check the circuit response.**Conditions :**

- Turn system enable switch OFF.
- Disconnect the rudder angle level sensor from the OEM sensor wiring harness.
- Turn system enable switch ON.

Action	Specification/Repair	Next Step

Wait 30 seconds. Check the ED-4 display to read the alarm codes.	Alarm Code 7266 or 7665 active and Alarm Code 7267 or 7664 inactive? YES	3D
	Alarm Code 7266 or 7665 active and Alarm Code 7267 or 7664 inactive? NO	4A

STEP 3D. Check the alarm codes and verify sensor condition.

Conditions :

- Turn system enable switch OFF.
- Disconnect the rudder angle level sensor from the OEM sensor wiring harness.
- Turn system enable switch ON.

Action	Specification/Repair	Next Step
Wait 30 seconds. Check the ED-4 display to read the alarm codes.	Alarm Code 7267 or 7664 active? YES Repair : A damaged sensor has been detected. See equipment manufacturer service information.	Repair complete.
	Alarm Code 7267 or 7664 active? NO Repair : None. The removal and installation of the connector corrected the fault.	Repair complete.

STEP 4. Check the original equipment manufacturer (OEM) sensor wiring harness.

STEP 4A. Inspect the OEM sensor wiring harness connector pins.

Conditions :

- Turn system enable switch OFF.
- Disconnect the OEM sensor wiring harness from the C.I.B.

Action	Specification/Repair	Next Step
• Loose connector • Corroded pins • Bent or broken pins • Pushed back or expanded pins • Moisture in or on the connector • Missing or damaged connector seals • Dirt or debris in or on the connector pins • Connector shell broken • Wire insulation damage • Damaged connector locking tab. Use the following procedure for general inspection techniques. Refer to Procedure 019-361 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-361.html)	Dirty or damaged pins? YES Repair : A damaged connection has been detected in the OEM sensor wiring harness connector. Clean the connector and pins. Repair the damaged harness, connector, or pins, if possible. Refer to Procedure 015-103 in Section 15. (/qs3/pubsys2/xml/en/procedures/513/513-015-103.html)	Repair complete.
	Dirty or damaged pins? NO	4B

STEP 4B. Check for a pin-to-pin short circuit in the OEM sensor wiring harness.

Conditions :

- Turn system enable switch OFF.
- Disconnect the OEM sensor wiring harness connector from the C.I.B.
- Disconnect the rudder angle level sensor from the OEM sensor wiring harness.

Action	Specification/Repair	Next Step
• Measure the resistance between the rudder angle level sensor SIGNAL pin in the OEM sensor wiring harness C.I.B. connector and all other pins in the OEM sensor wiring harness C.I.B. connector. Reference the circuit diagram or wiring diagram for connector pin identification. Use the following procedure for general resistance measurement techniques. Refer to Procedure 019-360 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-360.html)	Greater than 100k ohms? YES	4C
	Greater than 100k ohms? NO Repair : A pin-to-pin short circuit on the SIGNAL wire has been detected in the OEM sensor wiring harness. • Replace the OEM sensor wiring harness, if necessary. Refer to Procedure 015-103 in Section 15. (/qs3/pubsys2/xml/en/procedures/513/513-015-103.html)	Repair complete.

STEP 4C. Check for a pin-to-ground short circuit.**Conditions :**

- Turn system enable switch OFF.
- Disconnect the OEM sensor wiring harness connector from the C.I.B.
- Disconnect the rudder angle level sensor from the OEM sensor wiring harness.

Action	Specification/Repair	Next Step
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Measure the resistance between the rudder angle level sensor SIGNAL pin in the OEM sensor wiring harness C.I.B. connector and ground. Reference the circuit diagram or wiring diagram for connector pin identification. Use the following procedure for general resistance measurement techniques. Refer to Procedure 019-360 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-360.html)	Greater than 100k ohms? YES	4D
	Greater than 100k ohms? NO Repair : A pin-to-ground short circuit on the SIGNAL wire has been detected in the OEM sensor wiring harness. Troubleshoot each section of the harness and terminal block. Replace the OEM sensor wiring harness, if necessary. Refer to Procedure 015-103 in Section 15. (/qs3/pubsys2/xml/en/procedures/513/513-015-103.html)	Repair complete.

STEP 4D. Check for an inactive fault code.

Conditions :

- Connect all components.
- Turn system enable switch ON.

Action	Specification/Repair	Next Step

Wait 30 seconds. Check the ED-4 display to read the alarm codes.	Alarm Code 7267 or 7664 no longer active? YES Repair : None. The removal and installation of the connector corrected the fault.	Repair complete.
	Alarm Code 7267 or 7664 no longer active? NO Repair : A damaged sensor has been detected. Replace the rudder angle level sensor. See equipment manufacturer service information.	Repair complete.